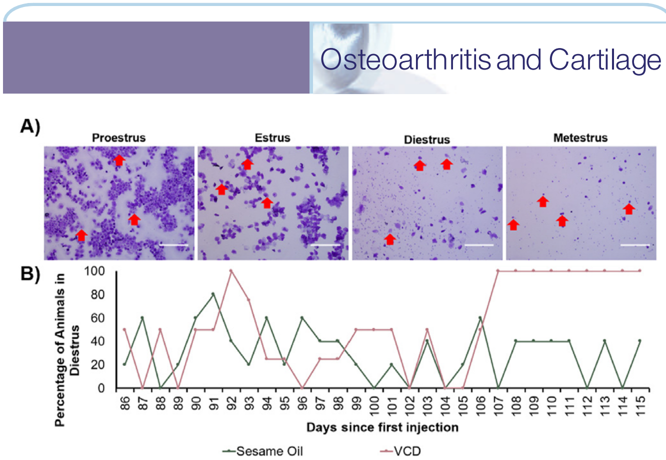

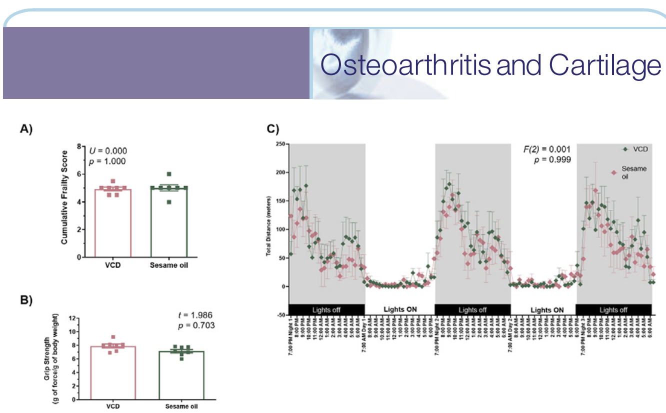

We saw no differences in subchondral bone grade across time in the menopause and non-menopause groups. There were also no differences between the menopause and non-menopause groups at each individual time point. (Figure 1D)

Conclusions: These findings suggest menopausal aging induces progressive cartilage degeneration in mice, while non-menopausal aging does not. Additionally, we found aging alone propagates increases in synovial pathology and menopause worsens these effects. Given the lack of differences in behavior and functional assays between menopausal and non-menopausal groups, we conclude these histological trends are not likely to be driven by menopause-induced changes in activity level.

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DESPITE UNIQUE WOUND HEALING ABILITIES, AFRICAN SPINY MICE (*ACOMYS*) ARE NOT PROTECTED FROM OSTEOARTHRITIS DEVELOPMENT FOLLOWING MENISCAL INJURY

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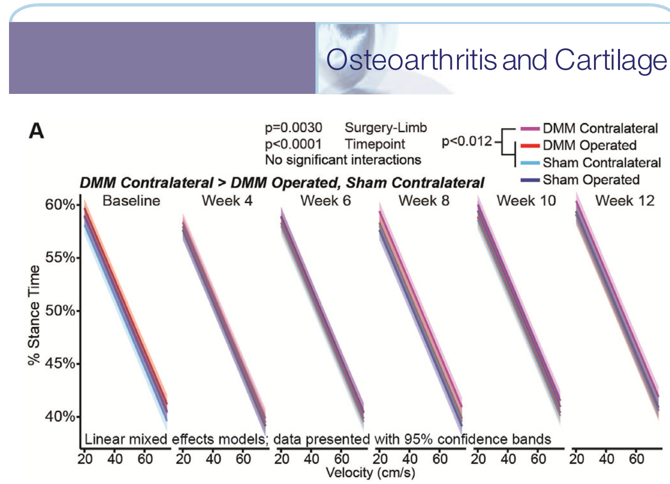
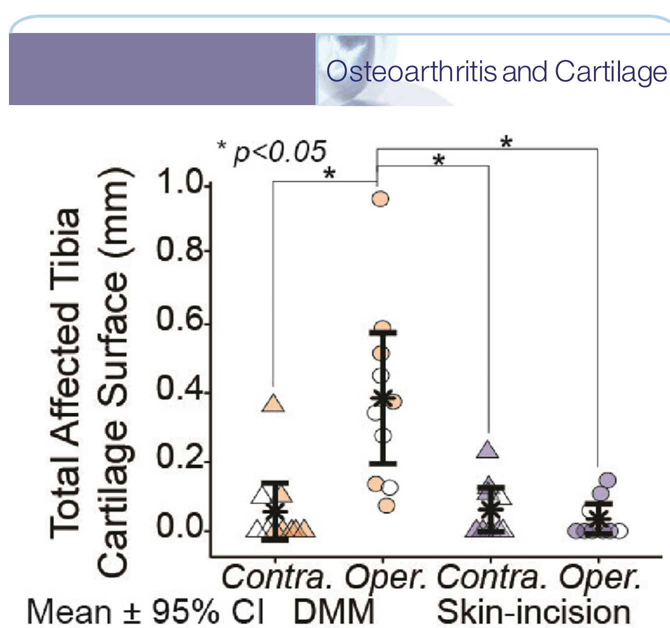
Purpose: African spiny mice (*Acomys*) have incredible wound healing abilities in skin, skeletal muscle, and elastic cartilage, demonstrating the ability to repair critically-sized defects that other mammalian species cannot. *Acomys* are an emerging species to study regenerative healing; however, it is unknown whether regeneration in *Acomys* extends to hyaline cartilage or whether *Acomys* are protected from osteoarthritis (OA) damage following joint injury. As such, this study evaluates if *Acomys* with meniscal injury are protected from developing OA-related joint damage or behaviors indicative of joint pain and dysfunction.

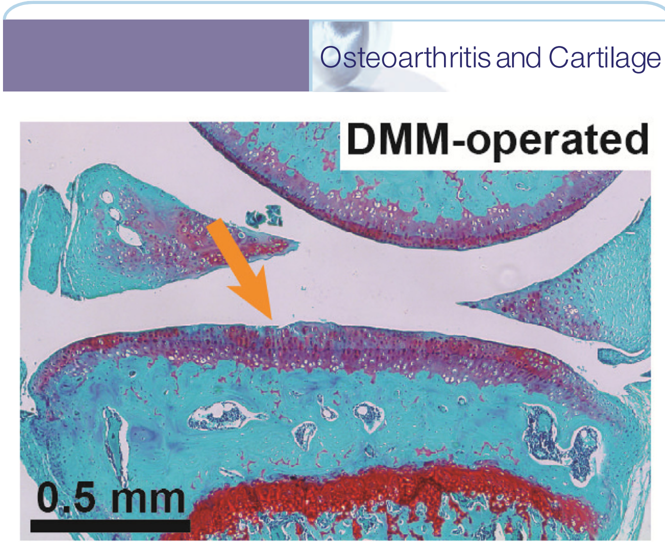
Methods: *Acomys* (n=14 male, n=7 female) received either a surgical destabilization of the medial meniscus (DMM) or a skin-incision sham surgery (n=11 DMM, n=10 sham, males and females evenly distributed). Spatiotemporal gait data were collected prior to surgery and at 4, 6, 8, 10, and 12 weeks post-surgery. At endpoint, semi-quantitative



Mankin grades and quantitative histological analysis of sagittal joint sections were used to assess joint damage in the medial compartment. Gait data were analyzed using linear mixed effects models, with fixed effects of surgery, timepoint, and velocity and the random effect of an animal identifier. Type III analysis of variance was used to analyze the significance of fixed effects (p<0.05) with Tukey's post-hoc comparisons of least squared means to evaluate specific group differences, when indicated. Nonparametric Mankin histology scores were analyzed with pairwise comparisons using the Mann Whitney U test and a Benjamini-Hochberg correction for multiple comparisons. Quantitative histological data were analyzed using a 1-way ANOVA and Tukey's HSD post-hoc test, when indicated. This study was approved by the University of Florida IACUC.

Results: *Acomys* with DMM altered their gaits by increasing the percent stance time on the uninjured, contralateral limb relative to the DMM-operated limb (p<0.012). This compensation increases the double limb support phase, thereby reducing the amount of time the injured limb bears weight on its own during a gait cycle. In contrast, skin-incision *Acomys* maintained balanced gaits, spending equal time on both hind limbs. Mild fissures along the medial tibial cartilage surface were observed in DMM-operated knees at 12 weeks after DMM surgery. These fissures coincided with higher Mankin sums in DMM-operated knees compared to contralateral and skin-incision controls (p=0.049). Reducing the Mankin sum to individual components showed structural damage of the tibial cartilage was the primary driver of group differences (p=0.0043); safranin-O staining did not differ between the





experimental groups. Quantitative histological analysis further verified the increased cartilage damage, with the total affected tibial cartilage surface being greater in DMM-operated knees compared to all other groups (p<0.0005).

Conclusions: Hyaline cartilage lacks innate healing abilities, and given *Acomys* unique healing abilities in muscle, skin, and elastic cartilage, we hypothesized that hyaline cartilage in *Acomys* would be protected from OA-related damage following DMM surgery. However, our data show *Acomys* did develop some histological evidence of OA following meniscal injury, and this damage coincided with mild gait compensations. While *Acomys* were not fully protected from hyaline cartilage damage, the severity of cartilage defects in DMM-operated *Acomys* was only mild to moderate, with none of the defects extending beyond the tidemark at 12 weeks post-injury. In contrast, DMM in C57BL/6 and 129/SvEv mice causes cartilage lesions that typically extend into the calcified cartilage by 8-10 weeks post-injury. As such, our histological findings could be interpreted as slowed OA progression in *Acomys*; however, this interpretation requires some restraint. *Acomys* are more closely related to gerbils than traditional laboratory mice; thus, conclusions related to the rate of OA progression should be tempered, as the rate OA progresses is known to vary by species. Moreover, gait data show DMM *Acomys* adopted a protective gait, indicating *Acomys* were also not protected from developing OA-related behaviors. Though we show *Acomys* are not fully protected from OA-related changes following a meniscal injury, *Acomys* may still be of interest to the orthopedic community, as the regenerative potential of other orthopedic tissues is currently under-explored and the rate of cartilage damage may be slowed relative to other rodents.

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SUBCHONDRAL BONE CHANGES IN OSTEOARTHRITIS ANIMAL MODELS AND HUMANS INDUCED BY MENISCAL LESIONS - A SYSTEMATIC REVIEW OF THE LITERATURE

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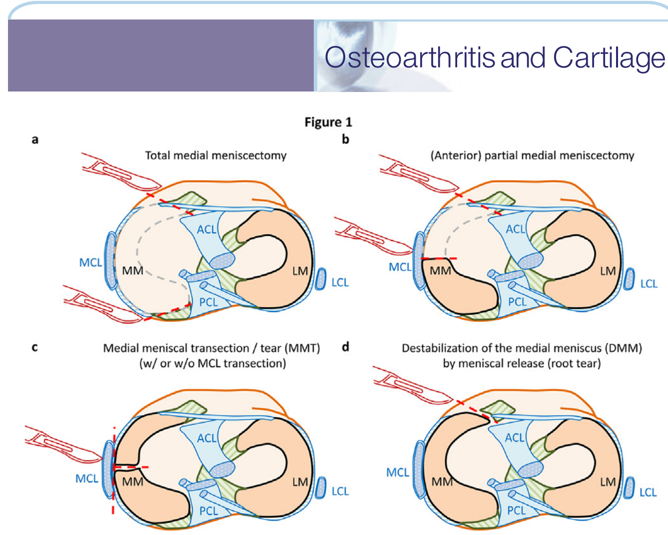
Purpose: Meniscal injuries lead to knee osteoarthritis (OA), which is also accompanied by macro- and microstructural alteration of the subchondral bone (SCB). Animal models are frequently used in OA research, however, to the best of our knowledge, no systematic reviews explored which techniques, species, and locations are used for experimentally modeling meniscal injuries. Such information could be useful to set up methodological standards for designing more comparable studies. Furthermore, the species- and surgical model-specific trajectory of SCB microstructural changes, and the fidelity of these models to reflect the human situation have not been evaluated yet.

Methods: A systematic PubMed search was performed (7/11/2022) using the terms "(osteoarthritis) AND ((meniscus) OR (meniscal) OR (meniscectomy))", retrieving 4998 results, including studies not reporting SCB data. Then, the search term "(subchondral bone) AND" was added, resulting in 471 papers, and after excluding papers reporting surgical combinations with anterior cruciate ligament transection or other techniques, n = 131 studies were included in the present systematic review.



Results: The 131 papers reported the structural consequences at the level of the SCB of 5 main types of meniscal damage: (1) non-surgical traumatic injuries (humans only), (2) total meniscectomy (TMX; removal of the entire meniscus), (3) partial meniscectomy (PMX; removal of a part of the meniscal tissue), (4) meniscal transection or (complete) tear (MMT; the mid-body of the meniscus is completely cut, but no parts are removed, animal models only), and (5) destabilization of the meniscus by meniscal release or root tear (DMM; the tibial attachment of the meniscus is completely transected, but no parts are removed) (Figure 1) in humans, sheep, minipigs, dogs, rabbits, guinea pigs, rats and mice. Altogether, 6.8% of the studies examined humans, 7.6% used large animals and 85.6% of them used small animals, mostly mice (59.8%) (Figure 2a). 61.4% of the studies used the right knee, and 20.5% of the studies did not report the knee where OA was induced. The majority of the studies induced OA in the medial compartment; however, almost 1/3 of the human studies did not report the compartment of the meniscal injury (Figure 2b). Traumatic meniscal tears were studied only in humans, TMX was used mainly in rabbits and larger species, PMX was studied in all species including humans, MMT was used in rodents and sheep and it was the most popular in rats, and DMM was used in all species except guinea pigs and was the most